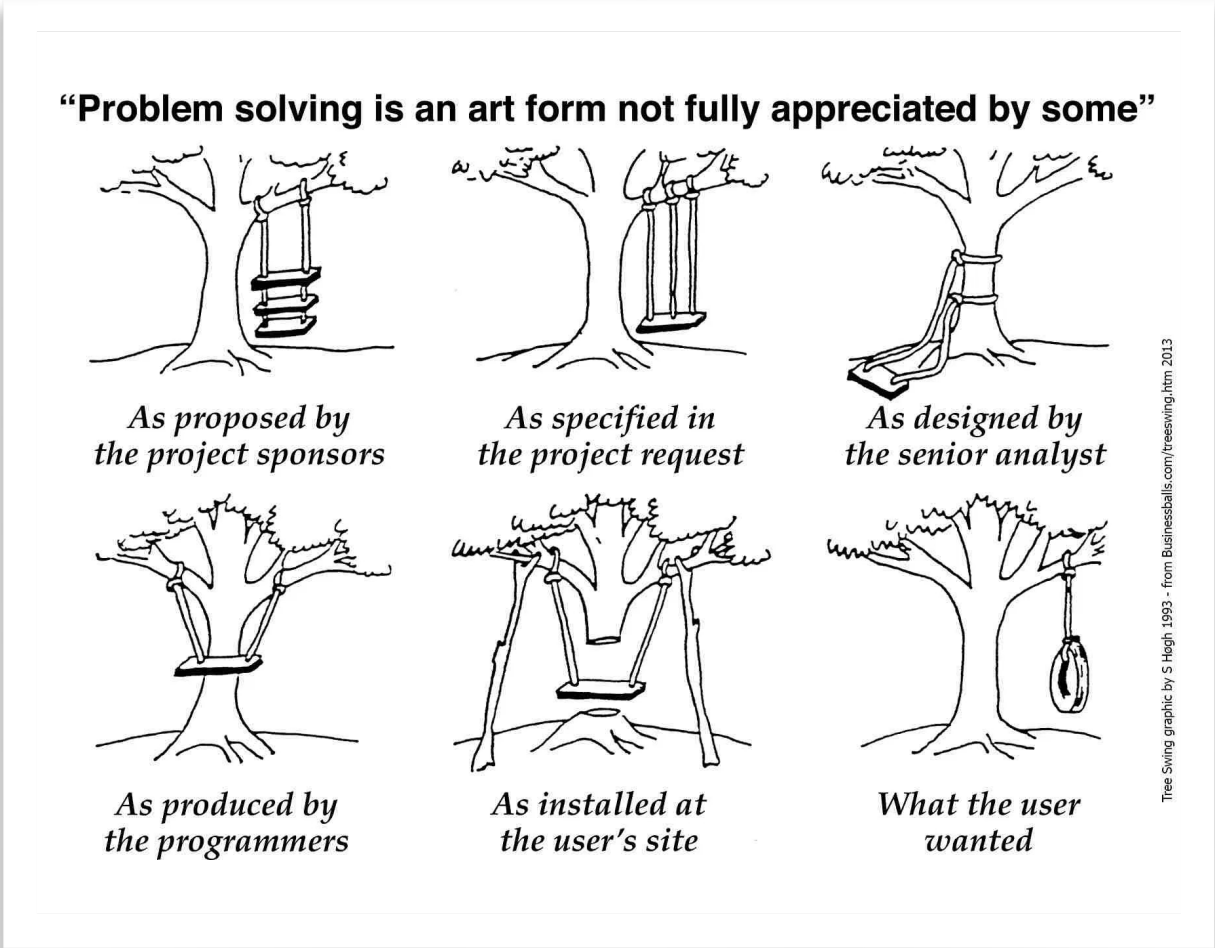
DECO3800/7380 Design Computing Studio 3 2024
Stephen Viller, School of EECS
Assignment 2-3 : Exhibit during class week 13 (Thursday 23 May, 2-6PM)
Documentation due during examination period (Monday 3 June, 1PM AEST)
Learning Objectives: 1, 2, 3, 4, 5, 6

Milestone 3: Exhibit



Tree Swing graphic (S. Høgh, 1993)

Project exhibit and documentation

Introduction

The final exhibit for the studio provides teams with the opportunity to showcase their proposed design in a public expo setting.

The exhibit is a social event where teams can present and discuss their creative technical work, which has developed through the semester as a well-considered solution to a

significant problem or challenge they have identified. The submitted documentation forms the primary piece of work for your project, providing a detailed description of project outcomes. In particular, it should clearly describe the problem identified, and the solution proposed. The project deliverables are the final iteration of the documentation and communication about the project that teams have been developing through the semester. They should be designed to form the starting point for a different studio team in DECO3801 Design Computing Studio 3 - Build, where teams finalise designs and build prototype solutions. Your design proposal will at least consist of the following:

- Project purpose
  - *What is your design intended to achieve? What positive impact will it make on the world?*
- Intended audience (stakeholders)
  - *Who will benefit--directly or indirectly--from the project you have designed?*
- How will your proposed design deliver it's outcomes to the stakeholders, and why have you designed it to be the way it is?
  - *This can be considered in terms of functional and non-functional requirements (e.g. the functionality and the user experience that delivers it)*
  - *It could be communicated through various means such as*
    - *Use cases (user journeys etc)*
    - *Design storyboards*
    - *Conceptual model*
  - Your design should be justified on the basis of the research you have conducted during the project (background and existing work, user studies, prototyping activities, etc.)

Each team will also need to produce a **1-page design brief** for their proposed project (see example in Blackboard). The brief will be used for all projects that are selected for Studio 3 - Build as the starting point for future teams selecting and deciding on their preferences for project selection.

You should not only include the proposed design but also identify success criteria and a process for how it should be evaluated. The design documentation will describe what is required to be developed, but not how. The technical detail of the design is the challenge for the next team in the Build part of Studio 3. You should pay particular attention to how the proposed design is appropriate for the problem context, including any organisational, cultural, or other situational factors, and a description of the ethical issues considered in the proposal, along with any specific measures taken to ensure that the design treats ethical matters appropriately.

This submission should also include any materials that would be used by a team charged with implementing the solution, e.g.details of user studies, data gathered through design activities such as the low/medium-fidelity prototypes and other items developed to test aspects of the proposed design.

Task Description

The Exhibit and Final Project Documentation checkpoint is the third and final deliverable of your studio project. It takes the form of a trade-show style public exhibit, and submission of

associated project documentation. Together, they communicate the outcomes of the team's design work this semester. From a holistic perspective, the documentation describes the team's proposed design, which is justified in terms of how it addresses the problem or design opportunity identified, how it is appropriate for this context—including meeting needs of any stakeholders—and is informed by evidence gathered through background research, evaluation of design ideas, and in response to feedback received.

- The documentation presents this information to an appropriate level of detail and clarity to allow a different team in Studio 3 - Build (DECO3801/DECO7381) to design and develop a solution on the basis of the documented work and outcomes. **Clear and concise communication is more important than the length of the documentation**. Non-relevant information should be removed.
- The exhibit will allow the team to present what they have done at a high level and in-person, where the team outlines the specific topic and justifies how their proposed design addresses it through explanation and discussion with the exhibit audience.
- As a final studio team deliverable, the documentation also specifically reports on how the team have addressed feedback from previous iterations of the design, and on their process and effectiveness at functioning as a team.

As was the case for previous iterations of the documentation, the deliverable should cover:

- A finalised description of the problem space you have identified and the background research and design exploration you have conducted through the semester. It is likely in most cases that you will have narrowed the scope to be more specific than what you started with.
- Key decisions you have made about the potential technology to be used, including alternatives considered and justifications for your choices (based on your research and design work).
- Consideration of the relative merits of your potential designs in terms of factors that will be meaningful and relevant to the stakeholders.
- A discussion of the risks to the overall project's success, and some techniques for mitigating these.
- Evidence of design exploration and engagement completed with stakeholders through sketching and prototyping aspects of the design to test ideas, and/or studies, interviews, etc. to gather stakeholder input and feedback.

These will be presented in the form of

- A trade-show style project exhibit, taking place during scheduled studio class in week 13. The exhibit will be publicised outside of the course and open to the public.
- Final documentation to be submitted via blackboard.

In keeping with how assessment has been designed for Studio 3, the criteria set out here are intended to provide you with support for what your exhibit and final documentation should focus on, and are not intended as a detailed specification for what you should prepare, as much of this will depend on the specifics of your project. The nature of the report as a 'living document' through the semester, with several iterations, means you are expected to be revising and supplementing what you have submitted at previous milestones rather than writing a new document. You may, of course, choose to restructure your documentation to change the focus from work-in-progress to final. This is more important if you have been given specific feedback on the structure of your previous submissions.

A reminder:

The task sheet for the first project milestone (Problem Identification) stated:

> *The end point for your project this semester will be a Proposal for a solution to the problem or opportunity you have identified, so as a team you should be undertaking activities that lead to a better understanding of what you are designing and why. The essential design challenge for your team, therefore, is to develop a well informed and justified set of requirements for a solution to the problem that you have identified.*

This means you are **not** expected to be building the solution, nor are you expected to be constructing an interactive prototype of the solution. You need to be focused on producing a proposal that clearly presents your design for a solution, which communicates what is needed for the desired solution, *not how to deliver or build* it[1].

The ultimate consumers of your final proposal documentation will be teams of students in DECO3801/DECO7831 Studio 3 - Build in the following semester. They will use the documentation as the starting point for their projects, where they will design and develop a solution based on the work you have done this semester (note: *further design work will almost always be required)*. From this perspective, the report needs to **clearly and concisely** communicate the outcomes of your work this semester, so that after reading your report the Build team have a clear understanding of the problem identified and the solution you have designed—*and why you have designed it to be like this*. It will be their job to design and build the specific prototype solution that conforms to the work you have done this semester, along with any further exploration they deem necessary. You should therefore stay focused on clearly communicating what is required, not on trying to specify the work of the build team.

Required content:

Your report and exhibit will be the final iteration of a design document which which has evolved as you have worked on it throughout the semester. At this stage, the main focus should be on finalising and clearly communicating your understanding of what your proposal describes, delivering a designed solution to a problem-in-the-world that is informed and justified according to the work that you have completed this semester. As with the previous iterations, you should ensure that this version of your report covers the same broad areas of content about the project. Further, as it is a final piece of documentation that will be handed over to a different team for further development, you should ensure it covers standard concepts such as intended use cases for the identified stakeholders, as well as the functional and non-functional (performance) requirements that should be met.

As was the case for the previous milestone, you should make sure you incorporate the ongoing feedback you have received by updating relevant content in the light of it. You should also include in the report an updated section that addresses how your team has functioned in terms of effective collaboration, including your strategy and approach to resolving or avoiding any teamwork problems encountered during the project.

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[1] see https://www.uxbooth.com/articles/build-better-requirements-documentation-why-who-and-how/ for more detail on this distinction

Due to the diverse nature of projects across the whole of Studio 3, each team will need to decide how to best communicate the required content in the report. The main requirements are unchanged from previous iterations of the report, and are not covered again here in the task description. What follows here relates to what needs to be considered differently in this final deliverable.

Final iteration of

In transitioning from a work-in-progress document to a final project milestone, you need to make sure that areas which previously were looking ahead towards further planned work and areas of uncertainty to be resolved now present the final position you have arrived at. This is not saying that you must have resolved all areas of uncertainty. It is saying you need to be clear about what you have finalised in your work and which further questions exist. Any such questions need to be presented as items for the build team to investigate, rather than future work for the current team.

- **Questions/areas of investigation**. While you should obviously aim to have resolved any questions previously identified, and tried to address all significant gaps in your knowledge about the problem and solution space, it is possible that some areas remain unresolved, particularly if they have arisen as a result of more recent design work. You should make sure these are communicated to the build team in a structured and prioritised manner so that they can quickly act at the start of next semester to resolve them.
- **Plan reconsidered as process followed**. It doesn't make sense to include a plan of future work in a final report. However, the plan from your previous deliverables is potentially a useful resource to explain your team's process and progress through the semester. It should therefore be rethought as part of your team's documentation and reporting of work done and the process followed.
- **Feedback**. As was the case for the Interim Critique milestone, you should make sure you have addressed how the project incorporated feedback received so far. As stated in the Interim Critique criteria:

  > *This isn't necessarily a separate section (although you could certainly include a high level summary to make sure that your audience's attention is drawn to key aspects of feedback and how you have responded to it). Feedback can come in many forms, from comments and questions after presentations, and on written work, through to less formal conversations and chats about your work. It is important that where feedback has led you to revise your thinking about the problem or your ideas for solutions, you should make sure that this is documented. It isn't necessary for you to have changed your project in response to every piece of feedback. You may, for example, have received feedback which contradicts the direction your work was going in, where that direction is well supported and justified by your background or user research. For non-trivial critique of this type, it is a good idea to justify/defend why you are explicitly not following such feedback (and why/based on which evidence) so that it is clear you have considered it.*

  It is also not necessary to list every single piece of feedback you have received and give a detailed point-by-point response to it. Clear communication of the requirements for a proposed solution on the basis of your team's design work should ultimately be goal of the documentation.
- **Team Reflection**. In addition to the existing report structure, each team should also include a commentary on the team's work on the project this semester. For a team to be effective, team members must pay attention to both the task they are collectively engaged in, and on the process of maintaining the team as an effective collaborative unit. The report should include a separate section which covers in particular reflection

on two broad areas: (i) *the content* - the direction that the project has taken and the team's work focused on the project goals, etc.; and (ii) on *the process* that the team has followed and how effectively they worked as a unit. Focus on both what worked well and what could have been done better.

Your task at this final stage of the semester continues to be to understand and describe the problem you are solving in more detail, so that you can propose a solution that is directly informed by this understanding, and justified in terms of your design exploration.

You should aim for the main body of the final report to be **15-20 pages (6000-8000 words).** This requirement is to ensure that you have taken care to communicate clearly and not add length to the document without carefully considering how it concisely communicates what you have done. Appendices can be included for ancillary material but the body of the report should be understandable without reading them. You will get a better outcome in terms of communicating your content (and therefore grade) if your report is clear and concise, with details that do not directly contribute to understanding the content being moved to appendices, or removed entirely. A report that is too long and therefore overwhelming for the reader is just as bad as one that is too short and missing detail. You are welcome to continue to format your report according to the template provided in the assignment folder on Blackboard, but this is still offered as a resource for you rather than a requirement for it to be used. It is suggested that this final iteration of the report is presented exactly as a coherent final report, rather than as a separate set of updates to the previous iteration of the report.

1-page

New requirement for 2024 is to produce a 1-page brief for your project to assist with potential hand-over to a different team of students in Studio 3 - Build. The purpose of the brief is to give a high level overview of a project, covering the following areas:

- Title
  - short descriptive title, along with a few #HashTag descriptors to assist with searching for similar projects.
- Project Brief
  - A high level description of what the proposed project should achieve--setting the technical and social context, and overall goals
- Problem Space
  - The context that the project takes place in. What is the problem or opportunity being addressed?
- Success Criteria
  - What must the finished project have to be considered a success?
- This Application MUST Have
  - Features that must exist in the final project
- This Application COULD Have
  - Optional features that could be included in the project
- Expected Skills
  - The set of skills that the team taking on the project will need to have or rapidly acquire

The Exhibit:

Rather than the kind of stand-up presentations we have used at previous project milestones in this course, the mechanism for presenting your final designs will take the form of a trade-show style exhibit. Each team should prepare a display of their work to be staffed by the team during the scheduled time for studio in week 13 of semester. Team members should

be prepared to talk about the project and discuss the design work you have done this semester to anyone who visits the exhibit. This should include a high-level 'elevator' pitch, as well as display materials to help further explain the problem and solution being addressed. There are no set requirements for what materials you should produce for the exhibit. The nature of the exhibit will be less formal than a stand-up presentation, so slide presentations are not appropriate, but posters to describe your project, example prototypes you have used to test your design ideas and gather user feedback, videos to illustrate the problem context, etc. are all relevant materials to use.

At the exhibit you should ensure that all team members are able to participate and contribute to the presentation of the semester's work, and all team members are able to describe and discuss any aspect of their team's project.

Submission

The final submission for the Project Proposal documentation will be via Blackboard. The submission will consist of the following items:

- The main report.
  - Prepared according to this task sheet. The main body of the report needs to be a document that can be understood in isolation of the other ancillary submitted items.
- 1-page brief
  - Prepared according to the guidelines in this task sheet and example in Blackboard.
- Appendices
  - Anything that is supplementary information for the report, but not vital for understanding the content. This may include interview data, sketches and other documentation of the design process, reports on studies carried out, tests/demos of technical aspects of the solution, etc. Many items here will be useful to the build team to provide supplementary detail or raw data/evidence that supports the design decisions you have made.
- Exhibit materials
  - Any materials produced as part of the exhibit display should also be submitted for review. This includes digital copies of any handout or poster material, and can also include links to interactive resources, photos of physical materials, or video material if this was included to help explain the project.

After assessment is finalised, the materials will be made available for students in the following semester. This will become a key resource for students taking Studio 3 - Build next semester in deciding which project to undertake as well as a resource for them to understand the proposed design and its justification

Criteria:

	EXCELLENT	SATISFACTORY	UNSATISFACTORY
"Depth of Treatment" (20%), "Required Content" (30%) and "Team Process Documentation & Reflection" (20%) are weighted higher than "Communication" (15%) and "Professionalism" (15%) in the overall grade, with the overall grade arrived at following a GPA-style calculation. The work should be developed to a standard such that it would be suitable to present to a potential investor, where the standard of production and professionalism of the team can influence a decision as strongly as the quality of the idea. Higher grades require higher quality of work, not quantity. Remember, these criteria are about the quality your work demonstrates and they are deliberately written in a way to allow you flexibility to fit them to your project context, rather than requiring you to change your project to fit them. If you are having problems understanding any criteria in the context of your project, please discuss with your tutor or with the teaching team more broadly. In keeping with the previous milestone, regarding how you have incorporated feedback so far, and a reflection on how you functioned as a team across the semester. The former will be embedded throughout the deliverable wherever feedback has had an impact on your work. Much of this should be obvious from how you present it anyway, but you may want to draw attention to significant impacts of feedback on your work to demonstrate how you have addressed it. The latter should be a separate section at the end of the report which presents a reflection from the team on the progress you have made this semester. This should address at least the following aspects with other topics at the discretion of the team. A reflection on the progress made compared against what you expected; and how effectively you functioned as a team.			
DEPTH OF TREATMENT (DOCUMENTATION)			
Background reading/ literature review (*what existing work has informed the project?*)	Of high quality showing evidence of wide research, extending well beyond the starting points identifying the problem or challenge	Adequate, but neither extensive nor exceptional	Shows signs of hurried or inadequate research
Creative work (*how much does the design reflect the team's work rather than reinventing the wheel? How well has background work and evidence from primary sources been used?*)	Original and creative synthesis of materials Sources are cleverly used to support the conceptual and technical work identified in the report	Adequate, but not original use of research materials Sources could be better used to support the proposal	Lacks originality Inadequate use of sources to support the concept and/or technology design

	EXCELLENT	SATISFACTORY	UNSATISFACTORY
Relevance to challenge or problem space (*how well does the proposed work address the identified problem?*)	Displays clear understanding of where proposed work fits into the bigger picture of the problem of challenge context	Shows occasional understanding of how the team's work is related to the context	Presents proposed work in isolation, with no connection at all to broader context
REQUIRED CONTENT (DOCUMENTATION)			
Stakeholders (*which people and/or organisations have a stake in the problem being solved?*)	Clear identification and justification of the intended stakeholders and how the project relates to them, and what use cases or scenarios the design should support.	Stakeholders are listed but not clearly related to rest of project. Use cases are included but superficially	Neither the stakeholders nor how the work relates to them are clearly defined
Conceptual model (*what is your final idea for the design of the solution? How should it be evaluated?*)	A high level description of the planned solution which clearly identifies the team's final intentions for the design, including functional and non-functional requirements and consideration of how they should be evaluated . Thought has been given to how this relates to background work and results from any study carried out so far to justify the design and requirements.	Idea is adequately defined, but not well justified or related to other work in the project.	No coherent description or justification of the idea

	EXCELLENT	SATISFACTORY	UNSATISFACTORY
Remaining questions/ uncertainty (*what areas of knowledge have you identified that still require investigation to learn more about the problem/solution?*)	Insightful points raised that demonstrate a good grasp of the team's understanding of the problem space; what, if any, further knowledge is required to improve understanding and advance the design process; and what activities will be required by the Build team to acquire this knowledge.	Basic questions identified with some thought given to how to answer them	Little effort shown to identify questions or how to satisfy them
Design activities (previously referred to as prototypes/ sketches/ studies) (*what you did to explore the solution space?*)	Activities are described which demonstrate how the team actively explored the problem and solution space, informed by the questions arising out of background research and creative design work to identify solutions. These activities may be sketches and prototypes to test ideas with stakeholders, planned studies for gathering data through interviews or focus groups or other ways of engaging stakeholders, as appropriate for the focus of the project. These activities have been cleverly designed and and used systematically as evidence to provide insights into the direction of the project and ultimately justify the design goals set and decisions made.	Activities are described which show that the team is considering stakeholder perspectives and input on the nature of the problem and the building towards a solution. Some effort has been made to design these activities to address questions that have arisen in the project, and there is evidence that the learnings from the activities are being used to inform the design.	Little or no activities have been undertaken to seek any input into the design of the project from outside of the immediate knowledge of the team members themselves. A survey or questionnaire might have been conducted but its design was not connected to relevant investigations for the project. It is difficult to understand how any of the design activities fit into a process of informing the proposed design from stakeholder input or knowledge of the problem context.

	EXCELLENT	SATISFACTORY	UNSATISFACTORY
Project Plan (re-presented as part of process documentation) (*how did you carry out your work as a team to achieve the project outcomes?*)	Clear breakdown of project into milestones, deliverables, strands, etc. as appropriate for the current final stage of the design, and documentation of any agreed processes for team maintenance.	Basic description of project plan, with minimal consideration for how the team worked effectively towards achieving it.	Basic task breakdown only and little/no evidence of any discussion about how the team members worked together.
Incorporation of feedback and new information (*how has the feedback you have received, and new knowledge gained through your design work, informed the project?)*	Feedback and critique from outside the team has clearly been actively sought and considered, and used as an opportunity to improve both the team's understanding of the problem, and their designs for a proposed solution. Where feedback calls for clarification, sections are updated accordingly. Where it calls for changes in direction, the changes are documented to explain and justify design decisions.	Feedback and critique from outside the team has been considered and incorporated into updates of the report and presentation.	The report and presentation provide no clear evidence that feedback has been appropriately responded to or incorporated into the team's work. OR it is clear that feedback has not been acted on or it has been actively ignored.
Ethical considerations *(How has the team addressed ethical concerns in relation to the topic of the project and the people who will be impacted by it?)*	Ethical considerations are central to the design process being followed and the team's understanding of ethics in relation to their project is a key driver for the development of their design. The team have actively used the ethical considerations for their project as a way to determine not only what they *should* be doing, but also what they *should not* do and why.	Ethical considerations have been listed and considered in an appropriate way for the project, but it is not clear how the team are incorporating them into the design.	Ethical concerns have not been identified or have been ignored. The project is progressing in a direction that may have a negative impact on the lives of the people who are being designed for

	EXCELLENT	SATISFACTORY	UNSATISFACTORY
Team process documentation and reflection	The team presents a reasoned and critical consideration of both task and process aspects of the project progress, providing insights into both strengths and weaknesses of the project and identifies key learning outcomes arising for the team. Problems arising from team conflict have been identified and addressed effectively, with or without support from the studio teaching team, but they have been fully aware of the situation and its resolution.	A good coverage of task and process aspects of project performance for the semester, but only superficial reflection. Insights are less convincing or harder to relate to the team's specific context. Problems have been identified, but maybe not fully resolved. Teaching team not fully aware of problems and help not sought in an effective manner.	No reflection or commentary is evident. The report includes a description of events and journal-style coverage of the semester without relating this back to what was learnt as a result and what conclusions can be drawn. Or, the section is used merely to air grievances rather than, for example, to look constructively at what can be done to overcome challenges. Problems not identified or addressed. Teaching team not advised, or advised too late in the semester to be able to help.
COMMUNICATION (EXHIBIT) The exhibit is a trade-show style event where teams will set up a display for the project and describe/discuss their work with visitors to the exhibit. It is not expected to (and should not be planned to) cover the full detail contained in the report, but present a high level 'pitch' of the project, while being able to provide further detail in response to questions and in discussion. Teams should focus on a clear description of their proposed solution, how it addresses the problem they have explored, and why it is a good solution (design justification). With the potential for the audience to include members of the public, teams should be prepared to describe the breadth and depth of the work they have done, and not rely on audience members having any prior knowledge of the problem or the technological solution being proposed.			
Communication design	Clear coherent flow of concepts and appropriate level of detail, easy to follow and understand, clearly relevant design for purpose	Good concepts and use of images/text/ materials, but not well related to challenge	No clear flow of concepts, resulting in a confused message; content either too high or too low level

	EXCELLENT	SATISFACTORY	UNSATISFACTORY
Verbal communication	Holds audience attention and conveys ideas strongly	Clearly audible, good flow and audience contact	Unclear and forced, little engagement, just read out slide contents
Design	The design of the display, visuals, etc. fit the context well and demonstrate flare in the team's approach to the problem	Exhibit design is appropriate for the content, but too much or too little use of images, etc.	Unclear how the exhibit design contributed to content, or design detracts from understanding the project
PROFESSIONALISM (DOCUMENTATION) This report and related documentation is the final report for the studio and should be delivered to a professional standard. The following criteria will now be applied for marking purposes, whereas previously you would have only received formative feedback to guide future work			
Language and Writing Style	Fluent and literate. Of high standard with good structure and flow and concise communication of content.	Adequate, but needs improvement. Keeps within bounds of word/ page count for body of report	Of low standard, hard to understand and follow meaning, too long, verbose wording that takes too long to read and understand
Referencing	Excellent support for the text and is of a consistent style throughout	Just adequate, or is not consistent style	Inadequate in number, or lack of consistent format
Proof Reading	No errors, clear that care has been taken to present a high quality report	Some errors have been missed, but well presented overall	Clearly not been checked for spelling, grammar, and typographical errors

Version information

Version 1.0	2/5/2024	Document created based on revision of the Final report assessment task sheet from 2023.	S.Viller